DARK STORE LOCATION OPTIMISATION

- Entities
  - I: Set of customer (demand) locations, indexed by i
  - J: Set of candidate dark store locations, indexed by j



- Parameters/Inputs
  - $Demand_i$: Daily Demand (number of orders) at customer location i
  - $Capacity_j$: Daily Capacity (max orders per day) of dark store at location j
  - $FixCost_j$: Fixed cost of opening dark store a location j
  - $DelCost_{ij}$: Cost of delivering per unit of shipment from dark store j to customer i

- Decision Variables:
  - $y_j \in \{0,1\}$: Binary variable to indicate if a dark store is opened at location j
  - $x_{ij} >= 0$. Share of daily demand of customer i served from dark store at location j

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- Constraint 1: Demand Satisfaction
  - $\text{Sum}_{j,\ \text{over all J}}\ x_{ij} = \text{Demand}_{i}$, for any customer i in set I

- Constraint 2: Facility Capacity
  - $\sum_{i \in I} x_{ij} <= \text{Capacity}_j\ y_j,$ $\quad \forall\ j \in J$

  - Observe the alternate ways in which symbols can be written (constraint 1 format and constraint 2 format)

- Objective
  - Minimize $\sum_{i \in I} \sum_{j \in J} \text{DelCost}_{ij} x_{ij} + \sum_{j \in J} \text{FixCost}_j y_j$

